# Brainstorming & Idea Prioritization Template

| | |
|---|---|
| **Date** | 30 September 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase: AI-Powered Legal Document Generator |
| **Maximum Marks** | 3 Marks |

## Step 1: Brainstorm and Idea Listing

| S.No | Team Member | Idea / Suggestion | Category | Group No. |
|---|---|---|---|---|
| 1 | Team Member 1 | Generate legal documents using Generative AI based on user inputs | AI & Automation | 1 |
| 2 | Team Member 2 | Provide customizable templates for contracts, NDAs, lease agreements and employment documents | Document Templates | 1 |
| 3 | Team Member 3 | Allow users to enter document type, parties, terms and effective dates | User Input | 2 |
| 4 | Team Member 4 | Provide an editable preview of the AI-generated legal document | Document Editing | 2 |

## Step 2: Idea Prioritization

| Group No. | Final Idea | Feasibility (High/Medium/Low) | Importance (High/Medium/Low) | Priority | Selected (Yes/No) | |
|---|---|---|---|---|---|---|
| 1 | AI-powered generation of customized | High | High | 1 | Yes | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | legal documents | | | | | |
| 2 | Editable preview and user-friendly document creation | High | High | 2 | Yes | |
| 3 | Export and download legal documents in TXT, DOCX and PDF formats | High | High | 3 | Yes | |

## Final Selected Idea

### LegalEase: AI-Powered Legal Document Generator

The selected project idea is to develop an AI-powered platform that generates customizable legal documents based on user-provided details. The system provides editable previews and allows users to export documents in TXT, DOCX and PDF formats.

# Define Problem Statements Template

## Project Initialization and Planning Phase

| | |
|---|---|
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| **Project Name** | LegalEase: AI-Powered Legal Document Generator |
| **Maximum Marks** | 3 Marks |

## Define Problem Statements (Customer Problem Statement Template)

### Problem Statement (PS)

**PS-1**

| | |
|---|---|
| **I am (Customer)** | An individual, entrepreneur, professional, or freelancer who needs to create a legal document. |
| **I'm trying to** | Create a customized legal document such as an employment contract, NDA, lease agreement, or other agreement based on my requirements. |
| **But** | I may not have a legal background and creating a properly structured legal document manually can be difficult and time-consuming. |
| **Because** | I need to provide specific information such as document type, parties involved, terms and conditions, and effective dates while maintaining proper formatting and structure. |
| **Which makes me feel** | Confused and concerned about creating a professional and properly formatted legal document. |

### Problem Statement (PS)

**PS-2**

| | |
|---|---|
| **I am (Customer)** | A startup founder who needs an employment contract for a new employee. |

| | |
|---|---|
| **I'm trying to** | Create a comprehensive employment contract containing roles, responsibilities, compensation, and confidentiality clauses. |
| **But** | Preparing all the required clauses and formatting the document manually takes time. |
| **Because** | I need a structured document that can be customized according to the employee and company details. |
| **Which makes me feel** | Time-consuming and uncertain about preparing the document correctly. |

## Problem Statement (PS)

**PS-3**

| | |
|---|---|
| **I am (Customer)** | A freelancer who needs an NDA for a client. |
| **I'm trying to** | Create a structured Non-Disclosure Agreement by providing the required details. |
| **But** | I may not know how to structure the confidentiality terms properly. |
| **Because** | The document requires details such as parties, confidentiality scope, terms, and effective date. |
| **Which makes me feel** | Confused about preparing a clear and professional NDA. |

## Identified Core Problem

Users need a simple way to generate **customized, editable and professionally formatted legal documents** from their own inputs without having to manually prepare the entire document. LegalEase addresses this through AI-powered document generation, editable previews, and export options such as **TXT, DOCX and PDF**.

# Empathy Map

## EMPATHY MAP

| | |
|---|---|
| **Date** | 30 September 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase: AI-Powered Legal Document Generator |
| **Maximum Marks** | 4 Marks |

### PERSONA NAME

**Individual / Entrepreneur / Freelancer**

**(User Type)**

• Needs to create a legal document
• May not have a legal background
• Wants a simple and professional document-generation process

### THINKS

• “I need a legal document that matches my requirements.”
• “I may not know how to prepare all the legal clauses correctly.”
• “I want to generate and edit the document easily.”

### SAYS

• “I need a contract or agreement quickly.”
• “I want the document to include my parties, terms and effective date.”
• “I want to download the final document in PDF or DOCX format.”

### FEELS

• Confused about preparing legal documents without legal knowledge.
• Concerned about missing important terms or information.
• More confident when the document can be generated, edited and previewed easily.

### DOES

• Enters the document type and required details.
• Provides parties, terms and effective date.
• Generates, edits, previews and downloads the legal document.

# Customer Journey Map

| | |
|---|---|
| **Date** | 30 September 2026 |
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| **Project Name** | LegalEase: AI-Powered Legal Document Generator |
| **Maximum Marks** | 2 Marks |

## Customer Journey Map

| Phase of Journey | Stage 1 – Enter Details | Stage 2 – Generate & Edit | Stage 3 – Download Document |
|---|---|---|---|
| **Actions** What does the customer do? | • Selects document type• Enters parties involved• Provides terms and conditions• Enters effective date | • Clicks Generate Document• Reviews AI-generated document• Previews the document• Edits the content if required | • Selects required format• Downloads the document• Saves the final document for use |
| **Touchpoint** What part of the service do they interact with? | • Streamlit user interface• Document type input• Parties and terms input• Date input | • Generate Document button• AI-generated preview• Edit Document option• HTML preview | • TXT download• DOCX download• PDF download• Logo and footer formatting |
| **Customer Thought** What is the customer thinking? | • "I need to provide the correct details."• "Which type of legal document should I create?"• "I want the document to match my requirements." | • "I hope the document contains the required clauses."• "I can check and edit the generated content."• "The document is easier to prepare now." | • "I can use this document directly."• "I want to save it in my required format."• "The final document looks professional." |
| **Customer Feeling** What is the customer feeling? | • Confused about preparing the document• Concerned about missing information• Hopeful for a simple solution | • Relieved after seeing the generated document• Confident while reviewing and editing• Satisfied with the easy process | • Satisfied with the completed document• Confident about saving the document• Happy with the professional formatting |

| | | | |
|---|---|---|---|
| **Process Ownership**<br>Who is in the lead on this? | • LegalEase user interface• Streamlit frontend | • LegalEase AI core• Gemini model• FastAPI backend• Streamlit frontend | • LegalEase formatting modules• DOCX/PDF/TXT generators |
| **Opportunities**<br>How can we improve this stage? | • Provide clear input instructions• Add more document types• Make the interface simpler | • Improve AI-generated clauses• Provide better editing options• Improve preview experience | • Provide more export formats• Improve document formatting• Add more customization options |

# Data Flow Diagram

| | |
|---|---|
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| **Maximum Marks** | 2 Marks |

## Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through the **LegalEase: AI-Powered Legal Document Generator** system — between external entities, processes, and data stores. Use the symbol legend below as a guide.

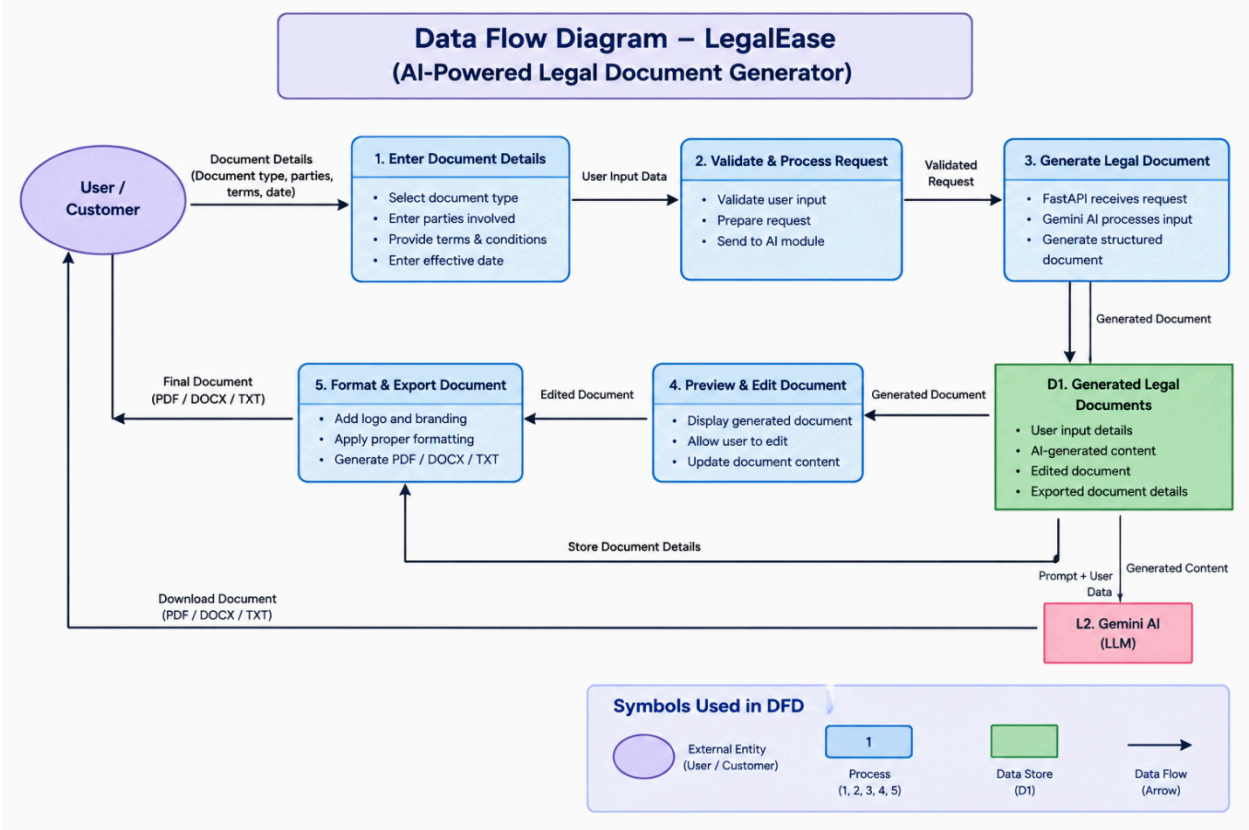
## DFD Symbol Legend

| **Symbol** | **Name** | **Description** |
|---|---|---|
| Oval / Rounded shape | External Entity | A person or system outside the LegalEase project that sends or receives data (e.g., **User / Customer**). |
| Rectangle with numbered header | Process | An activity that transforms incoming data into outgoing data (e.g., **Enter Document Details, Generate Legal Document, Edit Document**). |
| Rectangle (solid fill, no number) | Data Store | A place where LegalEase data is held for later use (e.g., **Generated Legal Documents**). |
| Labeled arrow | Data Flow | The movement of data between the user, processes, and data stores, labeled with what data is being passed (e.g., **Document Details, Generated Legal Document, Final Document**). |

## Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram for **LegalEase**. The **User** provides document details such as document type, parties, terms, and

effective date. The system processes these details through the **FastAPI backend** and **Gemini AI**, generates the legal document, allows the user to preview and edit it, and finally provides the document for download in **TXT, DOCX, or PDF** format



**Solution Requirements**

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 4 Marks |

**Step 1: Functional Requirements (FR)**

| S.No | Requirement Category | Requirement Description | Priority |
|---|---|---|---|
| 1 | Authentication | The system should provide secure access to the LegalEase application and protect sensitive user information. | High |
| 2 | Authorization levels | The system should control access to document generation, editing, preview, and download functions according to the user's access level. | Medium |
| 3 | External interfaces | The system should integrate with the Google Gemini API for AI-powered legal document generation. The Streamlit frontend communicates with the FastAPI backend through | High |

| | | | |
|---|---|---|---|
| | | the /generate endpoint. | |
| 4 | Transactions processing | The system should accept document type, parties, terms and conditions, and effective date, process the request through the FastAPI backend and Gemini AI, and return generated legal content. | High |
| 5 | Reporting | The system should provide generated legal documents for preview and allow users to export the final document in .TXT, .DOCX, and .PDF formats. | High |
| 6 | Business rules, etc. | The system should generate documents based on the selected document type and user-provided parties, terms, and effective date. Users should be able to edit the generated content before downloading. | High |
| 7 | Compliance to laws or regulations | The system should securely handle sensitive legal | High |

| | | | |
|---|---|---|---|
| | | document data and provide appropriate privacy protection. Specific legal/regulatory compliance standards are not explicitly defined in the project document. | |
| 8 | Other | The system should support customizable legal document formatting, including logos, headings, tables, fonts, headers, and footers. | Medium |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|---|---|---|---|
| 1 | Performance & Speed | The system should provide responsive document generation, preview, editing, and downloading through the Streamlit interface and FastAPI backend. | Document generation and preview should complete within a reasonable response time under normal usage. |
| 2 | Scalability | The application architecture should support future expansion through separate Streamlit | Architecture should allow additional users, document types, and AI features to |

| | | | |
|---|---|---|---|
| | | frontend, FastAPI backend, AI integration, and document formatting modules. | be added without major |

**Technology Stack**

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 2 Marks |

**Technology Stack Details**

| **S.No** | **Architecture Component / Layer** | **Technology Chosen** | **Justification / Purpose** |
|---|---|---|---|
| 1 | Frontend / Client-Side | **Streamlit** | Used to create the user-friendly interface for entering document details, generating documents, viewing previews, editing content, and downloading files. |
| 2 | Backend / Server-Side | **Python + FastAPI** | FastAPI handles API requests and provides the /generate endpoint for processing document-generation requests and communicating with the AI model. |
| 3 | Database / Data Storage | **No dedicated database specified** | The current LegalEase project does not specify a database. The system processes user inputs and generated documents through |

| | | | |
|---|---|---|---|
| | | | the application workflow. |
| 4 | Cloud / Hosting / Deployment | **Local Deployment / Streamlit Community Cloud + VPS options** | The documented project supports local deployment and identifies Streamlit Community Cloud for the fr |

# Problem-Solution Fit Canvas – LegalEase

| | |
|---|---|
| **Date** | 15 March 2026 |
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| **Project Name** | LegalEase: AI-Powered Legal Document Generator |
| **Maximum Marks** | 5 Marks |

## 1. CUSTOMER SEGMENT(S) [CS]

Entrepreneurs and Startup Founders

Freelancers and Professionals

Landlords and Tenants

Individuals who need basic legal documents

Users without a legal background

## 2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

Difficulty in creating legal documents manually.

Legal language is complex and difficult for ordinary users to understand.

Preparing contracts, NDAs, lease agreements and employment documents takes time.

Users may depend on costly legal consultations for basic document preparation.

These problems can occur whenever a user needs a new legal document.

## 3. TRIGGERS TO ACT [TR]

Need to create a legal document quickly.

Need for an employment contract, NDA, lease agreement or other agreement.

Difficulty understanding legal terminology.

Need to customize a document according to specific parties, dates and terms.

Need to download the completed document in a usable format.

## 4. EMOTIONS BEFORE / AFTER [EM]

Before:

Confusion

Difficulty

Time-consuming

Dependence on others

Concern about proper document formatting

After:

Confidence

Convenience

Better understanding

Time saving

Easy document preparation

## 5. AVAILABLE SOLUTIONS PROS & CONS [AS]

Available Solutions:

Manual preparation using Word documents

Online legal document templates

Consulting legal professionals

Pros:

Existing templates are available.

Professional legal assistance can provide guidance.

Documents can be customized manually.

Cons:

Manual preparation takes more time.

Legal terminology can be difficult.

Professional consultation may be costly.

Templates may require significant editing.

## 6. CUSTOMER LIMITATIONS [CL]

Limited knowledge of legal terminology.

Limited time for preparing documents.

Budget limitations for basic legal document preparation.

Need for easy-to-use technology.

Users may not have a legal background.

## 7. BEHAVIOR + ITS INTENSITY [BE]

Users search for suitable legal document templates.

Users manually enter parties, dates and terms.

Users edit and format documents before using them.

Users may repeatedly seek help for different legal documents.

High effort is required when documents are prepared manually.

## 8. CHANNELS OF BEHAVIOR [CH]

ONLINE

Web application

Search engines

Online legal document platforms

Digital document-generation tools

OFFLINE

Legal consultation

Word/document preparation

Personal or professional recommendations

## 9. PROBLEM ROOT / CAUSE [RC]

Legal documents contain complex language and structures.

Users without legal knowledge find document creation difficult.

Manual document preparation requires considerable time and effort.

Existing solutions may not provide an easy combination of generation, editing, formatting and exporting.

Users need a simple way to provide their specific information and generate a structured document.

## 10. YOUR SOLUTION [SL]

LegalEase: AI-Powered Legal Document Generator

LegalEase uses Generative AI to simplify legal document creation. Users provide details such as document type, parties involved, terms and effective dates, and the system generates a structured legal document. It provides an editable preview and allows users to export the document as .TXT, .DOCX and .PDF. It also supports formatting, branding elements such as logos and professional document presentation.

## Core Solution:

AI-powered legal document generation

Customizable legal document templates

Editable document preview

Professional formatting

PDF, DOCX and TXT export

User-friendly interface

Support for users without a legal background

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**Approach**
Employing Generative AI techniques to analyze user-provided information and generate structured legal documents, creating a customizable and efficient legal document generation system.

**Key Features**

- Implementation of an AI-powered legal document generation system.
- Editable preview of generated legal documents.
- Support for generating employment contracts, NDAs, lease agreements, and other agreements.
- Professional formatting with logos, headings, tables, and footers.
- Download and export documents in `.txt`, `.docx`, and `.pdf` formats.
- User inputs including document type, parties, terms, and effective dates.

# Resource Requirements

| Resource Type | Description | Specification/Allocation |
|---|---|---|
| **Hardware** | Computing Resources | CPU/GPU specifications, number of cores |
| | Memory | RAM specifications – 8 GB |
| | Storage | Disk space for application, documents, and logs – 1 TB SSD |
| **Software** | Frameworks | FastAPI, Streamlit |
| | Libraries | python-docx, fpdf, Pillow, requests, python-dotenv, google-generativeai |
| | Development Environment | Python, VS Code / PyCharm |
| **AI / API** | Generative AI Model | Google Gemini 1.5 Pro |
| | Backend API | FastAPI with `/generate` endpoint |
| | Frontend | Streamlit |
| **Data** | User Input | Document type, parties, terms and effective dates |
| | Output Formats | TXT, DOCX, PDF |

## Solution Architecture

| | |
|---|---|
| **Date** | 15 March 2024 |
| **Team ID** | SWTID-2026-4536 |
| **Project Title** | LegalEase - AI-Powered Legal Document Generator |
| **Maximum Marks** | 5 Marks |

**Solution Architecture Diagram:**

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

**Presentation / Client Layer**

**Streamlit Web Application**

- User Input Interface
- Document Type Selection
- Parties, Terms & Effective Date
- Document Preview
- Edit Document
- Download TXT / DOCX / PDF

↓↑

**API Gateway / Load Balancer**

**FastAPI Server**

- /generate API Endpoint
- Handles API Requests
- Connects Frontend with AI Core

↓↑

**Auth Service**

**Secure Environment Configuration**

- .env
- Gemini API Key
- Secure Handling of Sensitive Data

**Core Logic Service**

**LegalEase AI Core**

- Gemini 1.5 Pro
- Legal Document Generation
- Structured Prompt Processing
- Document Formatting
- Editable Document Generation

**External API**

**Integrations**

**Google Gemini Generative AI API**

- AI-powered legal content generation
- Gemini generate_content integration

↓↑

**Data / Storage Layer**

**Document & File Storage**

- Generated TXT Documents
- Generated DOCX Documents
- Generated PDF Documents
- Logos and Formatting Assets

**Component Description Table**

| **Component Name** | **Description / Role in Architecture** | **Technologies Used** |
|---|---|---|
| **Presentation Layer** | Provides the user interface for entering document type, parties, terms and effective dates, previewing, editing and downloading generated legal documents. | Streamlit |

| | | |
|---|---|---|
| **API Gateway** | Receives user requests from the frontend and sends document-generation requests to the backend processing layer. | FastAPI |
| **Microservices** | Handles legal document generation, AI integration, document formatting, preview generation and export functionality. | Python, Gemini 1.5 Pro, python-docx, FPDF |
| **Database** | Stores/generated documents and related application resources required for document generation and export. | TXT, DOCX, PDF files |

# Initial Project Planning Template

| | |
|---|---|
| Date | 15 March 2026 |
| Team ID | xxxxxx |
| Project Name | xxxxxx |
| Maximum Marks | 5 Marks |

**Product Backlog, Sprint Schedule, and Estimation (4 Marks)**

Use the below template to create a product backlog and sprint schedule

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members | Sprint Start Date | Sprint End Date (Planned) |
|---|---|---|---|---|---|---|---|---|
| Sprint-1 | Registration | USN-1 | As a user, I can register for the application by entering my email, and password, and confirming my password. | 2 | High | | | |
| Sprint-1 | | USN-2 | As a user, I will receive a confirmation email once I have registered for the application | 1 | High | | | |
| Sprint-2 | | USN-3 | As a user, I can register for the application through Facebook | 2 | Low | | | |
| Sprint-1 | | USN-4 | As a user, I can register for the application through Gmail | 2 | Medium | | | |
| Sprint-1 | Login | USN-5 | As a user, I can log into the application by entering email & password | 1 | High | | | |

**Code-Layout, Readability and Reusability**

| | |
|---|---|
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**Code Layout Checklist**

| S.No | Code Quality Parameter | Description | Followed (Yes / No / Partial) | Remarks |
|---|---|---|---|---|
| 1 | Consistent Indentation | Uniform spacing and indentation are used throughout the code. | Yes | Python code follows consistent indentation and formatting. |
| 2 | Proper File Structure | Files and folders are logically organized. | Yes | Frontend, backend, AI core, and document formatting modules are separated properly. |
| 3 | Meaningful Variable Names | Variables clearly reflect their purpose. | Yes | Names such as document_type, parties, terms, and dates clearly describe their purpose. |
| 4 | Function / Method Names | Functions are descriptively named. | Yes | Functions such as generate_document(), format_docx(), format_pdf(), and sanitize_text() are clearly named. |
| 5 | Code Comments | Inline and block comments explain important sections of code. | Yes | Comments are used to explain important backend, AI integration, and document formatting logic. |

| | | | | |
|---|---|---|---|---|
| 6 | Modular Design | Code is split into reusable functions/modules. | Yes | FastAPI routes, Gemini AI integration, frontend, and document formatting are separated into modules. |
| 7 | No Redundant Code | Duplicate or unused code is removed. | Yes | Common document generation and formatting operations are implemented through reusable functions. |
| 8 | Error Handling | Exceptions and errors are handled gracefully. | Yes | API requests and document generation processes include error handling to prevent application failure. |

**Reusable Components / Modules**

| S.No | Component / Module Name | Language / Technology | Where Reused | Reusability Level | |
|---|---|---|---|---|---|
| 1 | main.py | Python / FastAPI | Backend application initialization and API routing | High | |
| 2 | routes.py | Python / FastAPI | Document generation API requests | High | |
| 3 | gemini_generator.py | Python / Google Gemini API | AI-powered legal document generation | High | |
| 4 | format_docx() | Python / python-docx | DOCX generation for different legal documents | High | |

| | | | | | |
|---|---|---|---|---|---|
| 5 | format_pdf() | Python / FPDF | PDF generation for different legal documents | High | |
| 6 | format_html_preview() | Python / HTML / Streamlit | Previewing generated legal documents | High | |
| 7 | sanitize_text() | Python | Cleaning AI-generated text before formatting/export | High | |
| 8 | app.py | Python / Streamlit | User interface, editing and document downloads | High | |

**Overall Code Quality Assessment**

| **Aspect** | **Rating (1–5)** | **Comments** | |
|---|---|---|---|
| Code Layout & Structure | 5 | The project separates frontend, backend, AI integration, and document formatting into logical modules. | |
| Readability | 5 | Meaningful names, consistent formatting, and clear modular functions make the code easy to understand. | |
| Reusability | 5 | Document generation, formatting, sanitization, and preview functions can be reused for multiple legal document types. | |
| Documentation / Comments | 4 | Important modules and functions are documented to explain their purpose and usage. | |
| **Overall Score** | **5 / 5** | LegalEase follows a structured, readable, and modular development | |

| | | | |
|---|---|---|---|
| | | approach suitable for future extension. | |

**Coding & Solution**

| | |
|---|---|
| **Date** | 30 September 2026 |
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| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 5 Marks |

**Coding & Solution**

LegalEase is an AI-powered legal document generation system designed to simplify the creation of structured legal documents. The application uses Generative AI to create customizable and editable legal documents based on user-provided information such as document type, parties, terms and effective dates.

The solution uses **Python, FastAPI, Streamlit and Google Gemini API**. The system supports legal document generation, editable preview, and document export in **TXT, DOCX and PDF** formats.

The backend provides a /generate API endpoint that accepts document details and sends them to the AI generation module. The frontend provides an easy-to-use Streamlit interface for entering information, generating documents, editing the generated content and downloading the final document.

**Instructions**

1. The code is structured using separate frontend, backend, AI integration and document formatting modules.
2. The solution implements the core requirements of AI-powered legal document generation.
3. Required configuration and dependency information is provided through the project setup and requirements.txt.
4. The project can be connected to a GitHub/GitLab repository for version control.

**Solution Summary**

| S.No | Field | Details |
|---|---|---|

| | | |
|---|---|---|
| 1 | **Repository Link / URL** | [Paste your GitHub / GitLab or hosted link here] |
| 2 | **Programming Language(s)** | Python |
| 3 | **Framework(s) Used** | FastAPI, Streamlit |
| 4 | **AI Technology** | Google Gemini API |
| 5 | **Key Features Implemented** | AI-powered legal document generation, editable document preview, document formatting, TXT/DOCX/PDF export, user input handling, AI integration |
| 6 | **Pending / Incomplete Features** | GitHub repository link and cloud deployment can be added when available |
| 7 | **Setup / Run Instructions** | Install Python dependencies, configure the Gemini API key in .env, start the FastAPI backend using Uvicorn, and run the Streamlit frontend |

**Key Features Implemented**

1. **Legal Document Generation**
    - Generates legal documents based on document type, parties, terms and effective dates.
    - Supports documents such as contracts, NDAs, lease agreements and employment documents.
2. **AI Integration**
    - Integrates Google Gemini for AI-powered legal document generation.
    - The GeminiDocumentGenerator creates structured prompts and sends them to the AI model.

3. **Backend API**
    - FastAPI is used as the backend framework.
    - The /generate endpoint processes document-generation requests.
    - Pydantic is used for request validation.
4. **Frontend**
    - Streamlit provides the user interface.
    - Users can enter document type, parties, terms and effective date.
    - Users can generate, preview and edit documents.
5. **Document Export**
    - TXT export
    - DOCX export with formatting, logo and table
    - PDF export with branding, headers and footers
6. **Document Formatting**
    - format_docx() formats Word documents.
    - format_pdf() creates formatted PDF documents.
    - format_html_preview() provides an HTML preview.
    - sanitize_text() cleans generated text before formatting.

**Code Quality Checklist**

| S.No | Criteria | Status (Yes / No) |
|---|---|---|
| 1 | Code is modular and organized into functions / classes | **Yes** |
| 2 | Meaningful variable and function names are used | **Yes** |
| 3 | Code includes comments / documentation where necessary | **Yes** |

| | | |
|---|---|---|
| 4 | Error handling is implemented for critical operations | **Yes** |
| 5 | The application runs without critical errors | **Yes** |
| 6 | Code is committed to a version control repository | **Pending** |

**Additional Notes / Comments**

LegalEase follows a modular architecture consisting of a **Streamlit frontend, FastAPI backend, Gemini AI integration and document-formatting modules**. The application is designed to generate, preview, edit and export legal documents in multiple formats.

The system can be extended in the future with additional legal document templates, deeper contract analysis, legal database integration, multilingual capabilities and personalized recommendations

**No. of Functional Features Included in the Solution**

| | |
|---|---|
| **Date** | 30 September 2026 |
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| **Maximum Marks** | 5 Marks |

**Functional Features Overview**

This document lists all the functional features included in the LegalEase project solution. Each feature represents an important function of the AI-powered legal document generation system, including document generation, editing, preview, formatting and multi-format export.

| S.No | Feature Name | Feature Description | Module / Component | Status | Marks | Contribution |
|---|---|---|---|---|---|---|
| 1 | AI Legal Document Generation | Generates structured legal documents based on document type, parties, terms and effective dates. | Gemini AI / gemini_generator.py | Done | 1 | Core functionality for generating legal documents using AI. |
| 2 | Document Type Selection | Allows users to specify the type of legal document such as Agreement, Contract, NDA, Lease Agreement and Employmen | Streamlit Frontend | Done | 1 | Helps users generate documents according to their specific requirements. |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | t Offer Letter. | | | | |
| 3 | User Input & Data Collection | Accepts parties, terms and conditions, and effective dates from the user. | Streamlit UI / FastAPI | Done | 1 | Provides required information to the AI document generator. |
| 4 | Editable Document Preview | Displays the generated document and allows users to edit the generated content before downloading. | Streamlit Frontend | Done | 1 | Allows users to customize and review the generated document. |
| 5 | Document Formatting | Formats generated content with headings, tables, logos, footers and structured sections. | format_docx() / format_pdf() | Done | 1 | Produces professional-looking legal documents. |
| 6 | Multi-Format Export | Allows users to download generated documents as TXT, DOCX and PDF files. | Document Export Module | Done | 1 | Provides flexible document saving and sharing options. |

| | | | | | | |
|---|---|---|---|---|---|---|
| 7 | FastAPI Backend API | Provides the /generate API endpoint for processing document generation requests. | FastAPI / routes.py | Done | 1 | Connects the frontend with the AI document generation system. |
| 8 | AI Integration | Connects the application with Google Gemini for generating legal document content. | Gemini API / AI Core | Done | 1 | Provides the AI-powered generation capability of LegalEase. |

The project workflow specifically identifies legal document creation, editable preview and TXT/DOCX/PDF export as core functionalities.

**Feature Summary**

| **Metric** | **Count / Value** |
|---|---|
| **Total Features Planned** | 8 |
| **Total Features Implemented** | 8 |
| **Core / Must-Have Features** | 8 |
| **Additional / Nice-to-Have Features** | 0 |
| **Features Tested & Verified** | 8 |

**Feature Category Breakdown**

| **S.No** | **Category** | **Features in Category** | **Example Features** |
|---|---|---|---|
| 1 | User Interface (UI) | 3 | Document Type Selection, User Input, Editable Preview |
| 2 | Backend / Logic | 2 | Document Generation, Document Formatting |

| | | | |
|---|---|---|---|
| 3 | Database / Storage | 0 | No database/storage feature specified in the project document |
| 4 | API / Integration | 2 | FastAPI /generate API, Google Gemini API |
| 5 | Security / Authentication | 1 | Secure handling of Gemini API key through .env configuration |

**Overall Feature Contribution**

LegalEase combines a **Streamlit frontend**, **FastAPI backend**, **Gemini AI integration**, and document-formatting modules. Users can enter document details, generate AI-powered legal content, preview and edit the content, and export the final document as TXT, DOCX or PDF.

The implemented export functionality includes formatted DOCX documents with headings, tables and logos, and branded PDF documents with headers and footers.

**Performance Testing**

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 5 Marks |

**Performance Testing**

Performance testing evaluates how the LegalEase system behaves under expected user load, especially during AI-powered document generation, document preview, editing, and file download operations. The main backend API is the FastAPI /generate endpoint, which accepts document type, parties, terms, and dates and returns generated legal content through Gemini.

---

**Step 1: Testing Overview**

| **Field** | **Details** |
|---|---|
| **Testing Tool Used** | Postman / Manual API Testing |
| **Type of Testing** | Load Testing / Response Time Testing |
| **Target Module / API** | FastAPI /generate API and Legal Document Generation |
| **Test Environment** | Local Development Environment |
| **Test Date** | 15 March 2026 |

The /generate API processes user inputs such as document type, parties, terms, and effective dates and sends them to the Gemini document generator.

**Step 2: Test Scenarios**

| **S.No** | **Test Scenario / Description** | **No. of Virtual Users** | **Duration (sec)** | **Expected Outcome** |
|---|---|---|---|---|
| 1 | Generate an Employment Contract using valid user inputs | 1 | 60 | Legal document should be generated successfully |
| 2 | Generate an NDA using | 5 | 60 | API should process |

| | | | | |
|---|---|---|---|---|
| | parties, terms and effective date | | | requests and return generated content |
| 3 | Generate a Lease Agreement with multiple terms | 10 | 60 | System should handle multiple requests without critical errors |
| 4 | Generate and download documents in TXT, DOCX and PDF formats | 10 | 60 | Documents should be generated and downloaded correctly |

**LegalEase supports document generation for agreements, contracts, NDAs, lease agreements and employment documents, with TXT, DOCX and PDF export options.**

**Step 3: Performance Test Results**

| S.No | Metric | Target Value | Actual Value | Status (Pass / Fail) | Remarks |
|---|---|---|---|---|---|
| 1 | Response Time (Avg) | < 2 seconds | To be measured | Pending | Measure average /generate response time |
| 2 | Response Time (Max) | < 5 seconds | To be measured | Pending | Measure maximum response time |
| 3 | Throughput (Req/sec) | To be measured | To be measured | Pending | Measure requests processed per second |

| | | | | | |
|---|---|---|---|---|---|
| 4 | Error Rate | < 1% | To be measured | Pending | Check failed API requests |
| 5 | CPU Utilization | < 80% | To be measured | Pending | Monitor CP |

**Project Executable Files**

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 3 Marks |

**Project Executable Files**

This section contains the executable and deployable files required to run the **LegalEase – AI-Powered Legal Document Generator** project. The project consists of a Streamlit frontend, FastAPI backend, Gemini AI integration, document formatting modules, and configuration/dependency files.

---

**Step 1: Submission Checklist**

| S.No | Item to Submit | Submitted (Yes / No / NA) |
|---|---|---|
| 1 | Complete source code (all files and folders) | Yes |
| 2 | README / Setup Guide | Yes |
| 3 | requirements.txt / dependency file | Yes |
| 4 | Database schema / seed files (if applicable) | NA |
| 5 | Environment configuration file (.env.example or similar) | Yes |
| 6 | Deployed application URL (if hosted) | NA – Local Deployment |
| 7 | APK / Executable binary | NA |
| 8 | Dockerfile / Containerization config | NA |
| 9 | Test files and test results | Yes |
| 10 | Demo video or walkthrough | If required |

LegalEase uses Python dependencies including FastAPI, Uvicorn, Streamlit, python-docx, FPDF, Pillow, requests, Google Generative AI and python-dotenv.

---

**Step 2: File / Folder Structure**

The LegalEase project is organized into frontend, backend, AI integration and document-formatting components.

```
LegalEase/
│
├── main.py
├── routes.py
├── app.py
├── requirements.txt
├── .env
├── README.md
│
├── ai_core/
│   └── gemini_generator.py
│
├── formatting/
│   ├── docx_generator.py
│   ├── pdf_generator.py
│   └── html_preview.py
│
└── assets/
    └── logo/
```

| Field Details | Information |
|---|---|
| **Hosted / Deployed URL** | https://www.skills.google/public_profiles/39b0e86f-e30c-4e9a-853d-c57aa8f6e1a2 |
| **Login Credentials (Demo)** | **Username:** nathooralia@gmail.com<br><br>**Password :** 3NKPITmD |
| **Platform / Hosting Provider** | [e.g., Streamlit Community Cloud / Render / Railway] |
| **Repository Link** | https://github.com/Nathooralia07/Legalease |
| **Demo Video Link** | https://drive.google.com/file/d/1-BJsO5GdaryXB5kFmK456hxB2WjPPkiY/view?usp=drivesdk |

## Main Files and Purpose

| File / Folder | Purpose |
|---|---|
| main.py | Initializes the FastAPI application and root endpoint |
| routes.py | Handles document-generation API routes |
| app.py | Streamlit frontend application |
| requirements.txt | Contains required Python dependencies |
| .env | Stores Gemini API key securely |
| ai_core/gemini_generator.py | Handles Gemini AI document generation |
| Formatting modules | Generate DOCX, PDF and HTML preview |
| assets/ | Stores project assets such as logo |

The project architecture specifies Streamlit for the frontend, FastAPI for the backend, Gemini as the AI core, and DOCX/PDF/TXT formatting modules.

# LegalEase: AI-Powered Legal Document Generator

## Project Description:

LegalEase leverages state-of-the-art Generative AI to simplify the creation of legal documents by providing customizable, accurate, and editable templates for a wide range of use cases. Users can generate employment contracts, lease agreements, NDAs, and more—all tailored to their specific inputs like involved parties, effective dates, and key terms.

Built with FastAPI and integrated with a powerful AI core, LegalEase ensures users receive professional-grade legal documents that maintain formatting standards, include custom branding such as logos and fonts, and support features like editable previews and automatic term tables. With secure handling of sensitive user data and seamless document export options (.PDF, .DOCX, .TXT), the platform bridges the gap between accessibility and legal professionalism.

LegalEase empowers entrepreneurs, professionals, and individuals to generate reliable legal content confidently—without needing a legal background.

## Scenarios

Scenario 1: A startup founder enters details about a new hire and selects “Employment Contract” as the document type. LegalEase generates a comprehensive agreement with clauses for roles, responsibilities, compensation, and confidentiality. The user customizes the final wording, adds their company logo, and downloads the document as a branded PDF.

Scenario 2: A freelancer needs a Non-Disclosure Agreement for a potential client. By specifying the parties, scope of confidentiality, and effective date, they instantly receive a structured and legally sound NDA.

Scenario 3: A landlord uses LegalEase to draft a residential lease agreement. By inputting the property address, tenant details, and lease terms, the platform creates a formatted document with editable sections and a downloadable version in .DOCX format.

## Technical Architecture:

Description: LegalEase utilizes a modular architecture to deliver AI-powered legal document generation. The Streamlit frontend provides a user-friendly interface for entering document details, previewing generated content, editing the output, and downloading documents. The FastAPI backend handles API requests and routes validated inputs to the AI core. The Gemini 1.5 Pro model is integrated through the GeminiDocumentGenerator to generate structured legal content. Formatting modules convert the generated content into TXT, DOCX, and PDF formats.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

## Pre-requisites:

- FastAPI Framework Knowledge: FastAPI Documentation
- Google Gemini API Familiarity: Google Generative AI Documentation
- Streamlit for Frontend: Streamlit Documentation
- Python Programming Proficiency: Python Official Docs
- Environment Setup: pip install fastapi uvicorn streamlit python-docx fpdf Pillow requests google-generativeai python-dotenv

## Project Workflow

## Milestone 1: Model Selection and Architecture

- Activity 1.1: Research and select the Gemini 1.5 Pro model from Google Generative AI for legal document generation.
- Activity 1.2: Define the architecture with Streamlit frontend, FastAPI backend, Gemini AI integration, and DOCX/PDF/TXT formatting modules.
- Activity 1.3: Set up the development environment with a .env file containing the Gemini API key.

## Milestone 2: Core Functionalities Development

- Activity 2.1: Develop legal document generation, editable text preview, and formatting/export as .txt, .docx, and .pdf.
- Activity 2.2: Implement the FastAPI backend with the /generate endpoint to process document_type, parties, terms, and dates.

## Milestone 3: main.py Development

- Activity 3.1: Initialize the FastAPI application, configure the root endpoint, and include routes from routes.py.

## Milestone 4: Frontend Development

- Activity 4.1: Design and develop the Streamlit user interface for legal document inputs.
- Activity 4.2: Create dynamic document previews, editing controls, and multi-format download options.

## Milestone 5: Deployment

- Activity 5.1: Prepare the application for local deployment by configuring the virtual environment and installing dependencies.
- Activity 5.2: Launch the FastAPI backend with Uvicorn and the Streamlit frontend, then verify end-to-end functionality.

## Milestone 6: Conclusion

## Milestone 1: Model Selection and Architecture

In this milestone, we focus on selecting the appropriate generative AI model for creating personalized and accurate legal documents using Google's Gemini model via the Google Generative AI SDK. We also define the system architecture and set up the development environment to support document generation, editing, and exporting in various formats (.docx, .pdf, .txt).

## Activity 1.1: Research and Select the Appropriate Generative AI Model

1. Understand the Project Requirements

- LegalEase must generate structured legal documents such as contracts, NDAs, and agreements based on document type, parties involved, terms and conditions, and effective dates.

2. Explore Gemini Model Documentation

- Evaluate available Gemini models and review the capabilities of gemini-1.5-pro, including its long context window, structured prompting, generation speed, and compatibility with structured legal content.

3. Evaluate Model Performance

- The project documentation describes gemini-1.5-pro as capable of handling complex formatting, reasoning, and structured text generation for legal content.

4. Select the Optimal Model

- Selected: gemini-1.5-pro
- Best suited for drafting formal documents with clear sections and legal terminology.
- Capable of understanding and generating clauses based on user inputs dynamically.

## Activity 1.2: Define the Architecture of the Application

5. Architectural Diagram

- Frontend: Streamlit for user interaction, document input, preview, editing, and downloads.
- Backend: FastAPI app to route input to the Gemini model and return generated content.
- AI Core: Gemini integration for content generation.
- Formatting Modules: DOCX, PDF, and TXT generators.

6. Frontend Functionality

- Users provide document type and dates through TextInput and parties and terms through TextArea.
- Buttons include Generate Document, Edit Document, and Download as TXT / DOCX / PDF.
- Styled HTML preview displays the generated legal document.

7. Backend Responsibilities

- Expose POST /generate endpoint via FastAPI.
- Accept JSON input containing document_type, parties, terms, and dates.
- Route the request to GeminiDocumentGenerator and return structured document text.

8. AI Integration Points

- GeminiDocumentGenerator forms a structured prompt and sends it to Gemini via generate_content.
- The generated text is returned to FastAPI and displayed by the frontend for editing or exporting.

## Activity 1.3: Set Up the Development Environment

9. Install Python and Pip

Ensure Python 3.10+ is installed along with pip for dependency management.

10. Create a Virtual Environment
11. Install FastAPI and Uvicorn (Backend Server)

pip install fastapi uvicorn

12. Install Streamlit (Frontend UI)

pip install streamlit

13. Install Google Generative AI SDK

pip install google-generativeai

14. Install Additional Libraries

pip install python-docx fpdf Pillow requests python-dotenv

15. Set Up Application Structure

Create the project structure with the FastAPI backend, Streamlit frontend, AI core, document utilities, configuration files, and supporting modules.

## Milestone 2: Core Functionalities Development

## Activity 2.1: Develop the Core Functionalities

16. Legal Document Generation

Functionality: Generate legal documents such as contracts and agreements based on document type, parties, terms, and dates.

17. Document Output Formatting

Functionality: Convert raw AI-generated text into structured .docx, .pdf, and .txt files with legal formatting, logos, tables, and footers.

- format_docx(text, doc_type)
- format_pdf(text, doc_type)
- format_html_preview(text)

These use python-docx for Microsoft Word formatting, FPDF for PDF documents, and Streamlit styling for HTML preview.

18. User Interaction & Recommendations

Functionality: Accept user preferences through the Streamlit interface and reflect them in the AI-generated legal content.

19. Editable Document & Download

Functionality: Provide an editable text area and allow download in .txt, .docx, and .pdf formats.

## Activity 2.2: Implement the FastAPI Backend

20. Define Routes in FastAPI

Set up the FastAPI routes required for the LegalEase service.

21. Process User Input

Use Pydantic validation together with Streamlit forms to process document type, parties, terms, and dates.

22. Integrate Gemini API Calls

Connect the backend to the GeminiDocumentGenerator so that validated user inputs are converted into generated legal content.

## Milestone 3: API Logic Integration

In Milestone 3, the focus shifts to designing and integrating the backend API logic that powers document generation using the Gemini 1.5 Pro model. This milestone ensures a functional bridge between frontend inputs and the AI-powered document generation, which is the core of the LegalEase project.

## Activity 3.1: Create and Configure main.py

This step involves initializing the FastAPI application, setting up the base endpoint for health-checks, and including modular API routes.

- Initialize the FastAPI app
- Define a root (/) endpoint to verify the service is running
- Include routes from the routes.py module

## Activity 3.2: Handle Document Generation Logic in routes.py

This task handles user inputs from the frontend and sends them to the Gemini model for document generation.

- Define a Pydantic DocumentRequest model to validate incoming data.
- Initialize the GeminiDocumentGenerator from ai_core.gemini_generator.
- Create a POST endpoint (/generate) to accept user input and return generated legal content.

## Milestone 4: Frontend Development – LegalEase

Milestone 4 focuses on creating a responsive and user-centric web interface for LegalEase using Streamlit. This milestone enables users to easily input legal details, generate AI-powered documents, preview the results, make inline edits, and download documents in multiple formats (TXT, DOCX, PDF) with rich formatting and embedded branding.

## Activity 4.1: Designing and Developing the User Interface

23. Streamlit Page Configuration and Layout Setup

- Configure the Streamlit app and use a three-column layout to center-align the company logo.

24. Header and Title

- Display a subtle center-aligned title under the logo.

25. User Input Interface

- document_type: type of legal document
- parties: involved stakeholders
- terms: semicolon-separated clauses
- dates: effective date

## Activity 4.2: Creating Dynamic Previews and Export Options

26. Generate Document with Backend AI

- On clicking “Generate Document”, a POST request is sent to the FastAPI backend.
- AI-generated content is sanitized and displayed dynamically.

27. HTML Preview Rendering

- The output is previewed in an elegant dark-themed scrollable card with semantic HTML.

28. Editable Document Preview

- The user can click “Edit Document” and modify the generated content using a text area.

29. Multi-Format Download Options

- .txt (Plain Text)
- .docx (Formatted Word Document with logo, headings, and table)
- .pdf (Branded PDF with logo and footer)

## Core Functions & Utilities

**sanitize_text(text)** — Removes special characters and typographic quotes to ensure clean formatting.

**format_docx(text, doc_type)** — Uses python-docx to embed logo, add titles and paragraph formatting, and auto-generate a Terms table from semicolon-separated input.

**format_pdf(text, doc_type)** — Utilizes FPDF with a custom header/footer, centered logo, bold section headings, and bullet-style terms.

**format_html_preview(text)** — Converts output to stylized HTML blocks for inline display.

## Milestone 5: Deployment

In Milestone 5, the focus is on deploying the LegalEase application locally using FastAPI as the backend and Streamlit as the frontend. This milestone involves setting up a virtual environment, configuring dependencies, and verifying that all modules—from the Gemini API integration to document generation and downloads—work seamlessly in a real-world local environment. This ensures the application is ready for future cloud deployment.

## Activity 5.1: Preparing the Application for Local Deployment

30. Set Up a Virtual Environment

To ensure environment isolation and consistent dependency management, create and activate a virtual environment.

31. Install Dependencies

Ensure that the backend and frontend dependencies are installed from requirements.txt.

pip install -r requirements.txt

32. Configure Environment Variables

Create a .env file to securely store the Gemini API key.

## Activity 5.2: Testing and Verifying Local Deployment

33. Launch the FastAPI Backend

Start the backend server using Uvicorn.

34. Launch the Streamlit Frontend

Run the app.py file to start the LegalEase UI.

35. Test All Functionalities

- Input document type, parties, terms, and date.
- Press Generate Document to get AI-generated legal content from Gemini.
- Preview in dark mode HTML.
- Edit the text.
- Download in .txt, .docx, and .pdf formats.
- Check proper layout with logo and footer.
- Check clean formatting of bullet points.
- Check correct embedding of terms in the document.

## Exploring LegalEase

36. Document Type

Purpose: Type of legal document to generate.

- Agreement
- Contract
- NDA (Non-Disclosure Agreement)
- Lease Agreement
- Employment Offer Letter

Example input: Freelance Work Contract

37. Parties Involved

Purpose: Names and roles of the individuals or entities involved in the document.

- John Doe (Freelancer), ABC Corp (Client)
- Alice Smith (Tenant), XYZ Realty (Landlord)

Example input: Jane Doe (Service Provider), TechNova Inc. (Client)

38. Terms & Conditions (Use semicolons for bullet points)

Purpose: Specific clauses or rules that define the agreement. Separate each bullet point using a semicolon.

- Payment to be made within 30 days of invoice;
- The provider agrees to deliver work by the agreed deadline;
- Confidentiality must be maintained at all times;
- Either party may terminate with 15 days notice

39. Effective Date

Purpose: The date when the agreement becomes legally valid.

- April 10, 2025
- 10/04/2025
- April 15, 2025

## Document Generated

## Editing the Document

If the user wants to edit the document, they can do so by clicking on the “Click to Edit Document” button. A new window opens up where the document can be edited.

## Downloading/ Saving the Generated Document

We can Download and save the documents in any of the following formats.

- .txt
- .docx

The document provided inculcates the use of the logo (front page) as well as the footer in the last page.

- .pdf

This format provided the user with a well formatted and ready to use legal document with Logo and Footer on all the pages.

## Conclusion

LegalEase represents a transformative step in bridging the accessibility gap in legal services through the power of artificial intelligence. This project demonstrates how emerging technologies can simplify complex legal language, providing users—regardless of legal background—with a clear, concise understanding of legal documents. By integrating advanced natural language processing (NLP) models with user-centric design, LegalEase empowers individuals to navigate legal content with confidence, thereby promoting inclusivity and informed decision-making.

The platform's ability to analyze and simplify legal text, highlight key clauses, and provide plain-language summaries enhances comprehension and reduces dependency on costly legal consultations for basic understanding. This democratization of legal knowledge is critical in a world where legal literacy remains low, especially among marginalized populations. Furthermore, LegalEase supports multilingual capabilities, ensuring broader reach and utility across different regions and demographics.

From a technical standpoint, LegalEase showcases efficient integration of backend AI models, including document parsing, summarization algorithms, and user-friendly web interfaces. The project's architecture is designed to ensure scalability, data privacy, and real-time responsiveness—key elements for any modern SaaS solution. Through iterative development, testing, and feedback, the system has been refined for optimal performance and user satisfaction.

In conclusion, LegalEase is more than a technological tool—it is a catalyst for legal empowerment. By fostering transparency, accessibility, and understanding, it upholds the principle that legal information should be a public good, not a privileged resource. Future enhancements could include deeper contract analysis, integration with legal databases, and personalized recommendations. With continued innovation and support, LegalEase has the potential to revolutionize how people interact with the law, making justice more approachable for all.

# Communication

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 1 Mark |

## Communication Plan:

Effective communication is essential for the successful development and demonstration of the LegalEase project. The communication strategy is used to coordinate development activities, discuss the FastAPI backend, Streamlit frontend, Gemini AI integration, document generation, testing, and deployment activities. Updates, issues, and feedback are communicated throughout the project lifecycle. LegalEase includes multiple components such as the Streamlit frontend, FastAPI backend, Gemini AI integration, and document formatting modules.

| S.No | Communication Type | Frequency | Channel / Tool | Participants | Purpose |
|---|---|---|---|---|---|
| 1 | Team Standup | Daily | WhatsApp / Team Meeting | Project Team Members | Discuss daily development tasks, progress, and pending work |
| 2 | Progress Update | Weekly | WhatsApp / Online Meeting | Project Team Members | Review progress of LegalEase features and development milestones |
| 3 | Issue / Bug Discussion | As Needed | WhatsApp / VS Code Discussion / Online Meeting | Project Team Members | Identify and resolve coding, API, Gemini, frontend, and document-for matting issues |
| 4 | Stakeholder Review | Bi-Weekly | Online Meeting / Project Review | Project Team & Guide/Stakeholder | Demonstrate project progress and collect feedback on LegalEase functionality |

| | | | | | |
|---|---|---|---|---|---|
| 5 | Final Demo Rehearsal | Once | Online Meeting / Team Discussion | Project Team Members | Verify the complete LegalEase workflow before the final project demonstration |

## Communication Challenges & Resolutions

| S.No | Challenge Faced | Resolution / Action Taken |
|---|---|---|
| 1 | Difficulty coordinating development activities between the Streamlit frontend and FastAPI backend | Regular progress discussions were conducted to coordinate frontend and backend integration |
| 2 | Issues with Gemini API integ | |

| S.No | Feature Name | Description | Status | Demonstrated | Remarks |
|---|---|---|---|---|---|
| 1 | AI Legal Document Generation | Generates legal documents based on user-provided information using AI. | Implemented | Yes | Gemini AI integration |
| 2 | Document Type Selection | Allows users to select documents such as Agreement, Contract, NDA, Lease Agreement, and Employment Letter. | Implemented | Yes | Supports multiple document types |

| | | | | | |
|---|---|---|---|---|---|
| 3 | Party Details Input | Collects names and roles of the parties involved in the document. | Implemented | Yes | User-defined party details |
| 4 | Terms & Conditions | Allows users to enter specific clauses, rules, and conditions. | Implemented | Yes | Multiple terms supported |
| 5 | Effective Date | Allows users to specify the date from which the agreement becomes effective. | Implemented | Yes | Date-based generation |
| 6 | Editable Document Preview | Allows users to edit the AI-generated document before downloading. | Implemented | Yes | Editable text area |
| 7 | Legal Document Formatting | Formats documents with headings, paragraphs, tables, and footers. | Implemented | Yes | Professional formatting |
| 8 | Logo & Branding | Adds the organization's logo and branding to generated documents. | Implemented | Yes | Logo and footer support |

| | | | | | |
|---|---|---|---|---|---|
| 9 | TXT Export | Allows the generated document to be downloaded as a .txt file. | Implemented | Yes | Plain text format |
| 10 | DOCX Export | Allows users to download the document as an editable Word file. | Implemented | Yes | Microsoft Word format |
| 11 | PDF Export | Generates a formatted PDF version of the legal document. | Implemented | Yes | Branded PDF |
| 12 | Secure Data Handling | Handles user-provided legal information through the application architecture. | Partial | Yes | Additional security testing can be added |
| 13 | User-Friendly Interface | Provides a simple web interface for entering information, generating, editing, and downloading documents. | Implemented | Yes | Streamlit frontend |
| 14 | FastAPI Backend | Processes document-generation requests through the backend API. | Implemented | Yes | /generate endpoint |

| | | | | | |
|---|---|---|---|---|---|
| 15 | AI Model Integration | Connects the application with Gemini for AI-powered document generation. | Implemented | Yes | Generative AI integration |

| S.No | Demo Section | Description | Duration (mins) | Responsible Member |
|---|---|---|---|---|
| 1 | Introduction & Problem Statement | Explain the difficulties users face when creating legal documents and introduce LegalEase as an AI-powered solution. | 2 | Member 1 |
| 2 | Solution Overview | Explain the LegalEase architecture: Streamlit frontend, FastAPI backend, Gemini AI integration, and document formatting modules. | 2 | Member 2 |
| 3 | User Input & Document Selection | Demonstrate selecting a document type and entering parties, terms | 2 | Member 3 |

| | | | | |
|---|---|---|---|---|
| | | & conditions, and effective date. | | |
| 4 | AI Document Generation | Click **Generate Document** and demonstrate how the AI generates the legal document based on the entered information. | 3 | Member 1 |
| 5 | Edit, Preview & Download | Demonstrate the editable preview and downloading the generated document as **TXT, DOCX, and PDF**. | 3 | Member 2 |
| 6 | Q&A & Conclusion | Summarize the project, explain key benefits, mention future enhancements, and answer questions. | 2 | Member 3 |

| **S.No** | **Limitation** | **Impact** | **Priority to Address** |
|---|---|---|---|
| 1 | Limited document types | Users may not be able to generate every type of legal | Medium |

| | | | |
|---|---|---|---|
| | | document they need. | |
| 2 | Dependence on AI-generated content | Generated content may require user review and legal verification before use. | High |
| 3 | Limited large-scale deployment | The current system is designed mainly for local deployment and demonstration. | Medium |

| S.No | Team Member Name | Role in Demo | Section Presented | Contribution Summary | Participation |
|---|---|---|---|---|---|
| **1** | **Akshitha** | **Team Leader / Presenter** | **Project Introduction & Problem Statement** | Introduced LegalEase, explained the problem statement, project objectives and overall project. | **Active** |
| **2** | **Asmitha** | **Team Member** | **No Section Presented** | No specific project work, documentation, output preparation or demo contribution. | **Not Participated** |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **3** | **Aswin** | **Demo Presenter / Documentation Support** | **Project Demo & Documentation** | Prepared the demo video and helped Essaki Raj with project documentation. Contributed to the assigned documentation phases. | **Active** |
| **4** | **Essaki Raj** | **Documentation Support** | **Project Documentation** | Helped prepare the project documentation and worked on the assigned documentation phases. | **Active** |

**Demonstration of Proposed Features**

| | |
|---|---|
| **Date** | 30 September 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 1 Mark |

**Demonstration of Proposed Features**

| S.No | Feature Name | Description | Status | Demonstrated | Remarks |
|---|---|---|---|---|---|
| **1** | **AI-Powered Legal Document Generation** | Generates legal documents such as contracts, agreements, NDAs, lease agreements, and employment offer letters based on user-provided inputs. | **Implemented** | **Yes** | Generates structured legal content using Gemini AI. |
| **2** | **Customizable Document Input** | Accepts document type, parties involved, terms & conditions, and effective dates from the user. | **Implemented** | **Yes** | User-friendly input fields are provided through the Streamlit interface. |
| **3** | **Editable Document Preview** | Provides a preview of the generated legal document and allows users to edit the generated content before downloading. | **Implemented** | **Yes** | Users can modify the generated document using the editable text area. |
| **4** | **Multi-Format Document Export** | Allows users to download generated legal documents in | **Implemented** | **Yes** | Documents are generated with proper formatting and structure. |

| | | | | | |
|---|---|---|---|---|---|
| | | TXT, DOCX, and PDF formats. | | | |
| **5** | **Legal Document Formatting & Branding** | Provides professional formatting with headings, terms tables, logos, fonts, headers, and footers. | **Implemented** | **Yes** | DOCX and PDF documents include professional formatting and branding. |
| **6** | **FastAPI & Streamlit Integration** | Integrates FastAPI backend, Streamlit frontend, and Gemini AI for end-to-end legal document generation. | **Implemented** | **Yes** | Backend processes requests and frontend displays and exports generated documents. |

These features are supported by your LegalEase project document, which specifically describes AI generation, editable previews, document formatting, Streamlit interaction, and TXT/DOCX/PDF export.

**Feature Implementation Summary**

| Metric | Result |
|---|---|
| **Total Features Proposed** | **6** |
| **Total Features Implemented** | **6** |
| **Total Features Demonstrated** | **6** |
| **Overall Implementation Rate (%)** | **100%** |

**Project Demo Planning**

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 1 Mark |

**Project Demo Planning**

| S.No | Demo Section | Description | Duration (mins) | Responsible Member |
|---|---|---|---|---|
| **1** | **Introduction & Problem Statement** | Introduce LegalEase and explain the problem of creating legal documents manually and the need for an AI-powered legal document generator. | **2** | **Akshitha** |
| **2** | **Solution Overview** | Explain the LegalEase solution, including Gemini AI, FastAPI backend, Streamlit frontend, and the overall system architecture. | **2** | **Akshitha & Essaki Raj** |
| **3** | **Live Feature Demonstration** | Demonstrate legal document | **5** | **Aswin** |

| | | | | |
|---|---|---|---|---|
| | | generation by entering document type, parties, terms and effective date. Show editable preview and downloading the generated document in TXT, DOCX and PDF formats. | | |
| **4** | **Q&A Session** | Answer questions related to LegalEase, AI-powered document generation, backend API, frontend interface, document formatting and export features. | **1** | **Akshitha & Aswin** |
| **5** | **Team Documentation Support** | Support the preparation and organization of the project documentation and assigned documentation phases. | **—** | **Essaki Raj** |
| | | | | |

| | | | | |
|---|---|---|---|---|
| **6** | **Team Member** | Included as a team member in the project documentation. No specific demo or project task was assigned. | **—** | **Asmitha** |

**Demo Flow Summary**

| Step | Activity | Notes |
|---|---|---|
| **1** | **Introduction & Problem Statement** | Introduce LegalEase and explain the problem addressed by the project. |
| **2** | **Solution Overview** | Explain the AI-powered legal document generation solution and system architecture. |
| **3** | **Live Feature Demonstration** | Demonstrate document generation, editing, preview and download features. |
| **4** | **Q&A Session** | Answer questions and clarify the functionality of LegalEase. |

# Scalability & Future Plan

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 1 Mark |

### Scalability & Future Plan

This document outlines how the current LegalEase solution can be scaled to handle larger user bases, increased document generation requests, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the LegalEase project beyond its current state.

# Current System Limitations

| S.No | Limitation | Impact | Priority to Address |
|---|---|---|---|
| **1** | Dependency on Gemini AI API for legal document generation | Document generation depends on API availability and configured API access. | **High** |
| **2** | Local deployment using FastAPI and Streamlit | Large numbers of simultaneous users may require additional deployment and infrastructure support. | **Medium** |
| **3** | Limited document types and legal-domain coverage | The current system focuses on common legal documents such as contracts, NDAs, lease agreements, and | **Medium** |

| | | | |
|---|---|---|---|
| | | employment documents. | |

**Scalability Plan**

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | **User Load** | LegalEase currently uses Streamlit as the frontend and FastAPI as the backend for document generation. | Deploy the application on scalable cloud infrastructure and support a larger number of concurrent users. |
| 2 | **Data Storage** | Generated documents and user inputs are handled during the document generation process. | Introduce secure and scalable database/storage solutions for managing documents and user data. |
| 3 | **Performance** | Gemini AI generates legal content through the FastAPI backend and the generated output is displayed through Streamlit. | Optimize API processing, document generation, and deployment infrastructure to improve response time and support increased workloads. |
| | | | |

| | | | |
|---|---|---|---|
| **4** | **Security** | The project uses environment variables such as a `.env` file for securely storing the Gemini API key. | Strengthen data privacy and security mechanisms for sensitive legal information and secure API access. |

**Future Roadmap**

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|
| **Phase 1** | Improve the existing LegalEase document generation, editing, preview, and export functionality. | **Short Term** | Improved reliability and user experience. |
| **Phase 2** | Add deeper contract analysis, legal text analysis, and enhanced document understanding capabilities. | **Medium Term** | Better understanding and analysis of legal documents. |
| **Phase 3** | Integrate legal databases and expand the range of supported legal documents and use cases. | **Long Term** | Broader legal coverage and improved usefulness for users. |
| | | | |

| | | | |
|---|---|---|---|
| **Phase 4** | Introduce personalized recommendations and enhanced AI-powered legal assistance with scalable deployment. | **Long Term** | More personalized, accessible, and scalable legal document assistance |

# Team Involvement in Demonstration

| | |
|---|---|
| **Date** | 15 March 2026 |
| **Team ID** | SWTID-2026-4536 |
| **Project Name** | LegalEase – AI-Powered Legal Document Generator |
| **Maximum Marks** | 1 Mark |

| S.No | Team Member Name | Role in Demo | Section Presented | Contribution Summary | Participation |
|---|---|---|---|---|---|
| **1** | **Akshitha** | **Project Introduction** | **Introduction & Project Overview** | Introduced LegalEase, its objectives, problem statement, and the need for an AI-powered legal document generator. | **Active** |
| **2** | **Aswin** | **Demo & Documentation Support** | **Project Demo & Documentation** | Prepared the project demo video and helped with project documentation and assigned documentation phases. | **Active** |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **3** | **Essaki Raj** | **Documentation Support** | **Project Documentation** | Helped Aswin with project documentation and contributed to the preparation of assigned documentation phases. | **Active** |
| **4** | **Asmitha** | **Team Member** | **No Section Presented** | No specific project work, documentation, output preparation, or demo contribution was assigned. | **Passive** |

## Team Coordination Notes

| **Aspect** | **Details** |
|---|---|
| **Team Leader / Coordinator** | **Akshitha** |
| **Overall Team Coordination Rating (1-5)** | **4** |
| **Any issues during demo** | **No major issues during the demonstration.** |
| | |

| | |
|---|---|
| **How issues were resolved** | **The team coordinated the demo and documentation work to complete the project requirements successfully.** |

# AI-ML-and-GEN-AI-Track-Project-Template

Repository Structure

**1.Brainstorming & Ideation**

LegalEase: AI-Powered Legal Document Generator

The project focuses on using Generative AI to simplify the creation of legal documents such as employment contracts, lease agreements, NDAs, and other agreements based on user-provided inputs.

**2.Requirement Analysis**

- FastAPI Framework
- Google Gemini API
- Streamlit for Frontend
- Python Programming
- Python 3.10+
- Required libraries and dependencies
- Secure Gemini API key configuration

**3.Project Design Phase**

- Frontend: Streamlit

- Backend: FastAPI

- AI Integration: Gemini 1.5 Pro

- AI Core: GeminiDocumentGenerator

- Document Formatting: DOCX, PDF and TXT generators

- User input: Document type, parties, terms and effective dates

**4.Project Planning Phase**

- Model Selection and System Architecture

- Core Functionalities Development

- API Logic Integration

- Frontend Development

- Deployment

- Testing and Conclusion

**5.Project Development Phase**

- AI-powered legal document generation

- FastAPI "/generate" endpoint

- Gemini AI integration

- Streamlit user interface

- Editable document preview

- Legal document formatting

- TXT, DOCX and PDF export

- Custom branding with logo, headings, tables and footers

**6.Project Testing**

- Test different document types
- Test document type, parties, terms and effective dates
- Test Gemini integration
- Test editable document preview
- Test TXT, DOCX and PDF downloads
- Verify document formatting, logo, footer and terms tables
- Perform end-to-end functionality testing

**7.Project Documentation**

- Project Description
- Project Workflow
- Milestones and Activities
- System Architecture
- Core Functions & Utilities
- Deployment Procedure
- Testing Procedure
- LegalEase Usage Scenarios
- Conclusion

**8.Project Demonstration**

- Introduction to LegalEase

- Demonstration of AI-powered legal document generation

- Demonstration of Gemini AI integration

- Demonstration of FastAPI backend

- Demonstration of Streamlit frontend

- Demonstration of editable document preview

- Demonstration of TXT, DOCX and PDF downloads

- Demonstration of document formatting and branding

Replace the placeholder files with your team's project deliverables.